

Supplemental materials

Supplemental Movies

In all movies, individual fish follow data-inferred behavioral rules coupled to interactions with flows and domain boundary. Parameter values correspond to the cases highlighted Figs. 1 and 2 of the main text.

Movie S.1: Swarming versus Elongated Milling.

Movie S.2: Milling versus Double Milling.

Movie S.3: Polarized Schooling, Following the Tank Wall, and Turning.

Movie S.4: Fish Speed Increases with Increasing School Size.

Movie S.5: Bistability versus Intermittent Bistability.

Supplemental Figures

Eleven supplemental figures (Figs. S.1–S.11), one supplemental table (Table S.1), and one supplemental algorithm (Algorithm S.1).

Fig. S.1: Model set-up.

Fig. S.2: Statistical order parameters associated with Fig. 1.

Figs. S.3 and S.4: Comparison of classic and new statistical order parameters that distinguish between the collective patterns in Fig. 1.

Fig. S.5: Statistical order parameters associated with Fig. 2.

Figs. S.6: Increasing average speed with increasing school size.

Figs. S.7, S.8, S.9, and S.10: Effects of changing school size N , wall aversion I_w , domain size R , and domain shape on the intermittent behavior in Fig. 2.

Fig. S.11: Same phase diagrams as in Fig. 3 for a smaller school of size $N = 10$.

Supplemental Information

Formulation of the mathematical model and equations of motion.

Table S.1: We used the behavioral rules proposed in [18] with parameter values inferred from experimental data on groups of barred flagtails (*Kuhlia mugil*) in a shallow tank, with typical fish length about 10 cm, and typical swimming speed about 2 bodylengths per second, $U = 0.2$ m/s. The tank diameter in these experiments was 4 m. To obtain non-dimensional counterparts, we used the characteristic length $U^{1/2}k_p^{-1/2} = 0.7$ m and characteristic time $(Uk_p)^{-1/2} = 3.5$ s. With this choice, a non-dimensional tank radius $R = 10$ is nearly 70 times the fish bodylength.

	Attraction Intensity	Alignment Intensity	Wall Avoidance	Fish Speed	Fish Length	Tank Radius
Experiment [18]	k_p	k_v	k_w	U	ℓ	R
	0.41 $\text{m}^{-1} \text{s}^{-1}$	0.27 m^{-1}	0.19 m s^{-1}	0.2 m s^{-1}	0.1 m	2 m
Nondimensional	I_p	I_a	I_w	U	ℓ	R
	1	0.20	0.94	1	0.14	2.85
Ranges in model	1	0.1–10	0.5–4	1	0.14	4–10

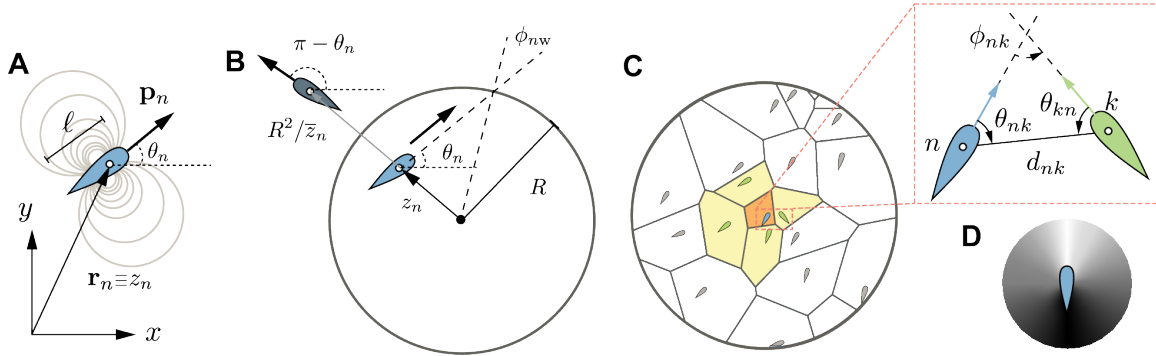


Figure S.1: **Model Set-up.** **A.** Far-field signature of a swimming fish in an unbounded domain corresponds to a dipole singularity located at z_n and oriented at an angle θ_n from the inertial x -direction. In vector form, the fish position is given by $\mathbf{r}_n = x_n \mathbf{i} + y_n \mathbf{j}$ and its heading is described by the unit vector $\mathbf{p}_n = \cos \theta_n \mathbf{i} + \sin \theta_n \mathbf{j}$. **B.** Image system of single fish in circular confinement consists of a 'ghost' dipole located at $R^2/\bar{z}_n$ outside the circular domain and oriented at $\pi - \theta_n$. The fish responds visually to the wall depending on its distance d_{nw} and heading angle ϕ_{nw} from the normal to the wall. **C.** Fish uses vision-based rules to respond to its Voronoi neighbors (highlighted in yellow). These vision-based rules are based on rules of attraction and alignment with Voronoi neighbors in response to the parameters explained in the pairwise context in the inset. **D.** A fish modulates its response to its Voronoi neighbors by a continuous visual field, shown in grey scale from white (perfect vision) to black (no vision).

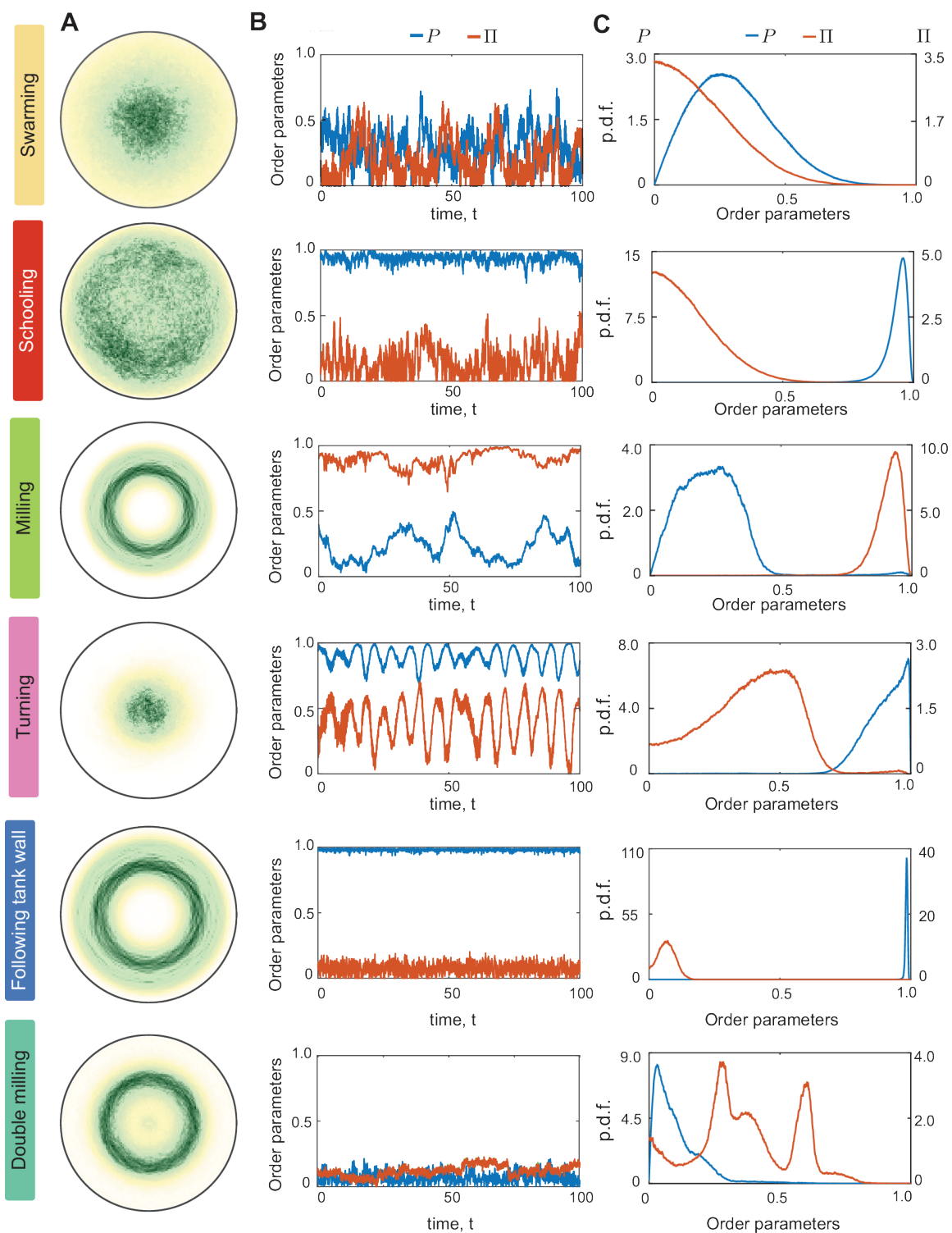


Figure S.2: **Statistical analysis of emergent collective phases shown in Fig. 1 of the main text:** **A.** Heatmap of locations visited by all fish over time (darker green correspond to more frequent visits). **B.** Evolution of polar and rotational order parameters. **C.** Normalized probability density functions (histograms) of polar and rotational order parameters based on 100 random realizations of the fish initial conditions. Parameter values for swarming $I_n = 0.8$, $I_p = 0.3$; schooling: $I_n = 0.3$, $I_p = 6.5$; milling: $I_n = 0.3$, $I_p = 1.5$; turning: $I_n = 0.1$, $I_p = 4.0$; following tank wall: $I_n = 0.1$, $I_p = 7$; double milling: $I_n = 0.1$, $I_p = 0.2$. In all cases, $N = 100$ and $R = 10$ except in double milling $R = 4$. Integration time is 400 time units, only the last 100 times units are shown in panel B.

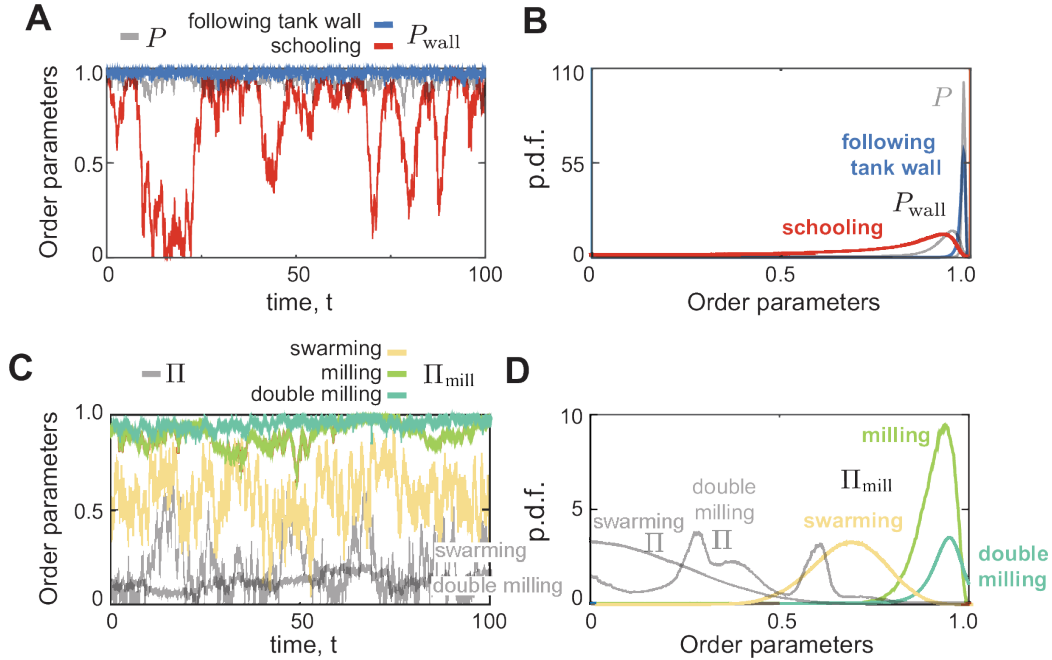


Figure S.3: **Distinction between schooling and following the tank wall and between swarming, milling, double milling:** Time evolution of P_{wall} for the schooling and following the tank wall; P superimposed in grey for comparison. When following the tank wall, P_{wall} stays consistently near unity for all time, while when schooling P_{wall} varies significantly. **B.** Normalized p.d.f.'s of P_{wall} for 100 realizations. When schooling, the p.d.f. has a wide variance while when following the tank wall, it is narrowly supported near unity. The p.d.f.'s of P are superimposed for comparison. **C.** Time evolution of Π_{mill} for swarming, milling, and double milling; Π superimposed in grey for comparison. While Π makes no distinction between swarming and double milling, Π_{mill} does. **D.** Normalized probability density functions of Π_{mill} based on 100 random realizations. The p.d.f.'s of Π are superimposed for comparison. Parameter values correspond to those shown in Fig. S.2 and Fig. 1 of the main text

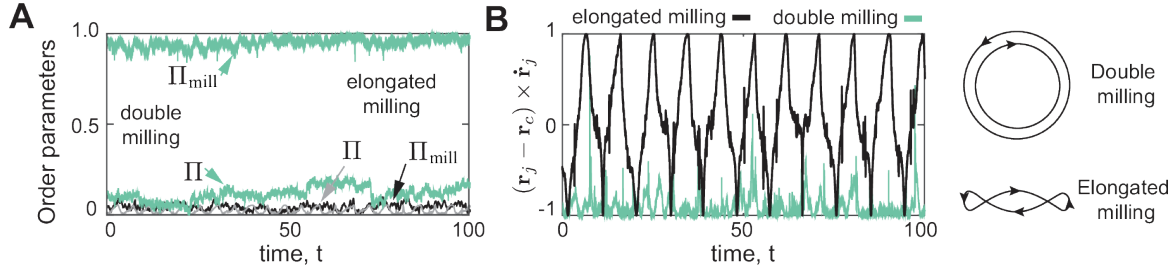


Figure S.4: Distinction between double milling and elongated milling: **A.** Time evolution of Π and Π_{mill} for double milling and elongated milling. **B.** Time evolution of the angular momentum $(\mathbf{r}_j - \mathbf{r}_c) \times \dot{\mathbf{r}}_j$ of a single fish, chosen randomly, shows that in elongated milling, individual fish periodically switch their direction of rotation, while in double milling, a fish rarely switches direction. Parameter values correspond to those shown in Fig. 1 of the main text

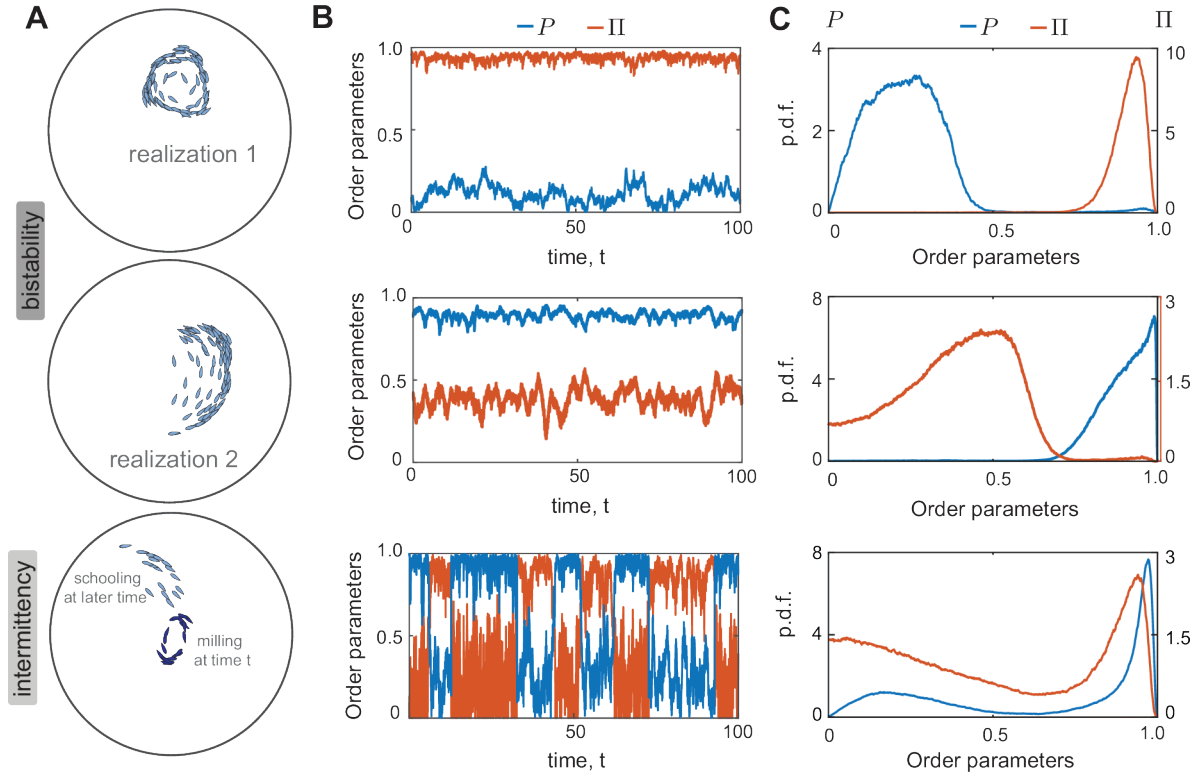


Figure S.5: Bistability: **A.** Snapshots of the fish school taken from two realizations corresponding to different initial conditions but the same parameter values. **B.** Time evolution of polar and rotational order parameters. **C.** Normalized probability density functions (histograms) of polar and rotational order parameters based on 100 random realizations. Parameter values are: $N = 100$, $I_n = 0.1$, $I_p = 3.0$. **Intermittency:** **A.** Snapshot of fish school at two instants in time. In this case, **B.** Time evolution of polar and rotational order parameters. **C.** Normalized probability density functions (histograms) of polar and rotational order parameters based on 100 random realizations. Parameter values are: $N = 20$, $I_n = 0.3$, $I_p = 1.2$

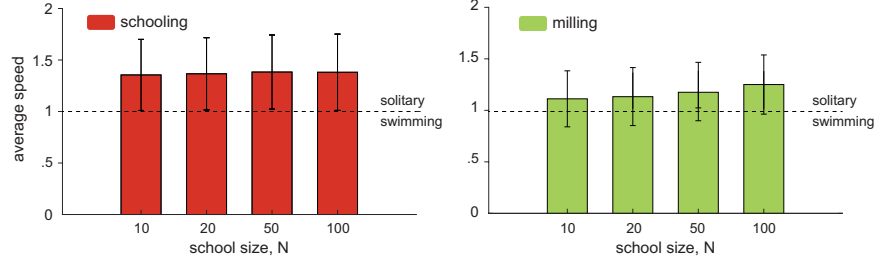


Figure S.6: Increase in average speed with school size: In the ordered states – schooling, milling, turning, and following the tank wall– flow interactions causes the average fish speed $\sum_{j=1}^N \|\dot{\mathbf{r}}_j\|/N$ to increase compared to solitary swimming, which we normalized to $U = 1$. Importantly, here we show that the average speed also increases with increasing group size N , thus recapitulating the experimental observations that average group speed positively correlates with local order [30] and school size [9]. In all cases, $I_a = 1$, $I_n = 0.3$, $R = 10$.

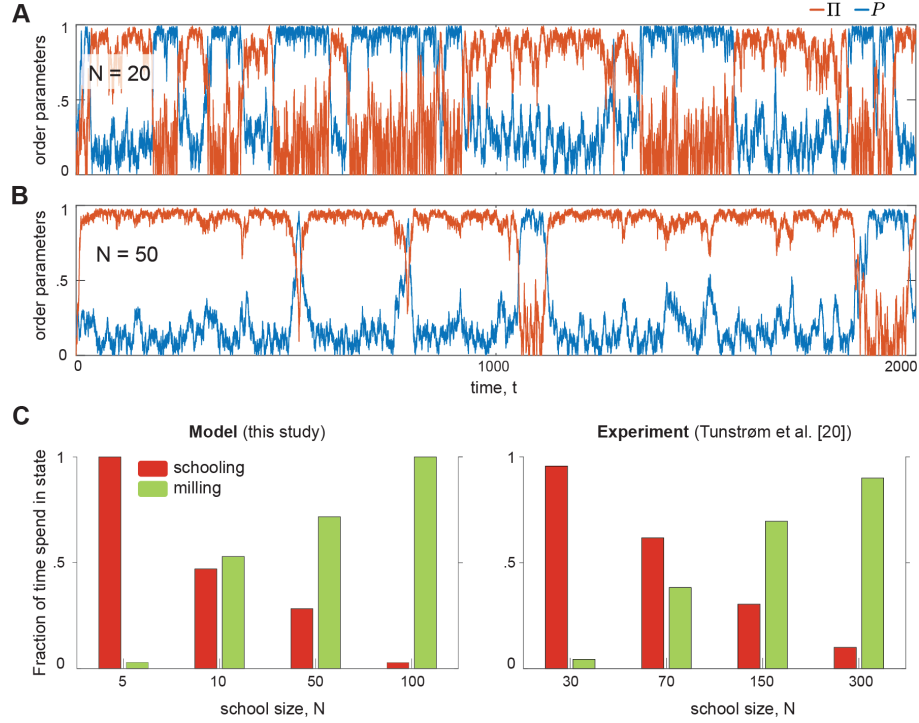


Figure S.7: Intermittent bistability: Switching rate between schooling and milling decreases with increasing group size. Time series of P and Π shows that **A** a group of 20 fish exhibits higher switching rate than **B** a group of 50 fish. **C** Fraction of time spent in each of the milling and schooling states as a function of school size N . To calculate this time fraction, we integrated the equations of motion for a total time period $T = 1000$, discarded the transient dynamics to ensure steady state is reached. Model results reproduce qualitatively experimental observations of confined fish groups in [9, Fig. 4A]. Note that in [9, Fig. 2], the authors define the thresholds between swarming, milling and schooling differently than the definition adopted here and in [21]. Thus, we scaled their data in [9, Fig. 4A] for consistency. In all cases, $I_a = 0.8$, $I_n = 0.3$. $R = 10$.

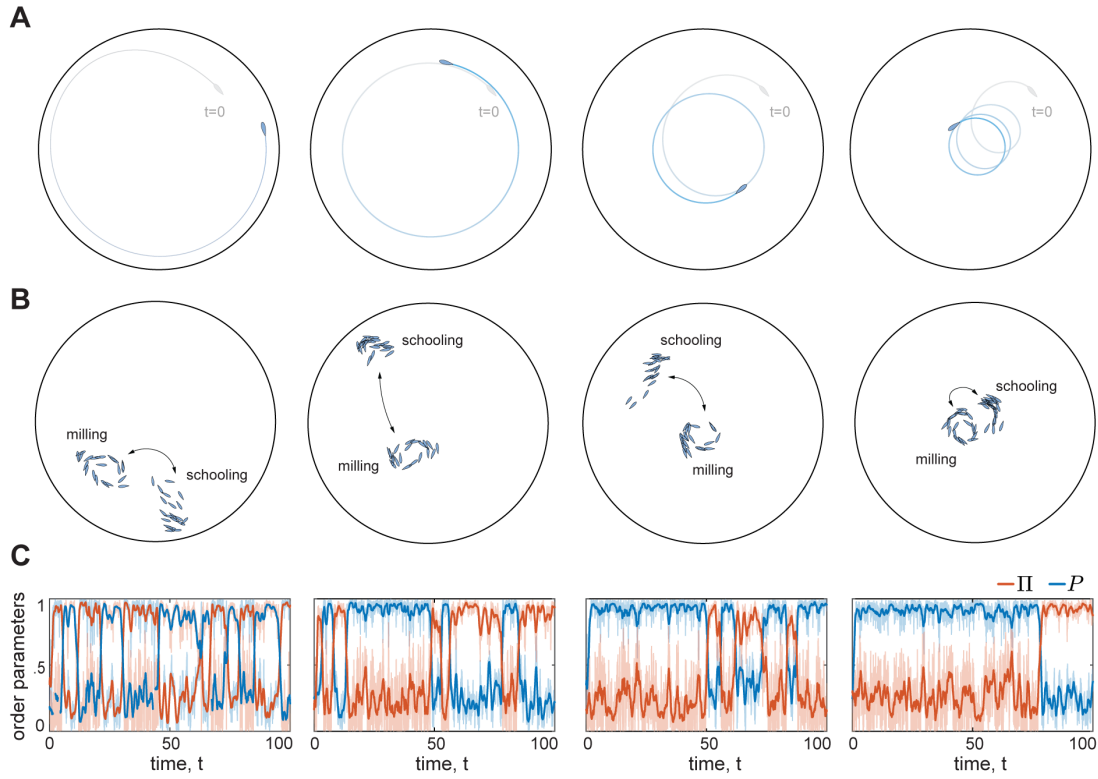


Figure S.8: Interaction with boundary affects intermittent bistability: **A** Behavior of a single fish as I_w is increased from 0.94 to 4×0.94 , all starting from the same initial conditions. As I_w increases, the fish "sees" and responds to the tank wall from farther distances, thus limiting its range of motion in the tank. **B** a group of $N = 20$ fish exhibits intermittent milling and schooling for all values of I_w , however, the switching rate between the two states depends inversely on I_w with higher rate at smaller I_w . **C** This is evident from the time evolution of the order parameters, here depicted for a time period of 100 units selected after steady-state was reached. These results indicate that interactions with boundaries have prominent effects the switching dynamics. In all cases, $I_a = 1$, $I_n = 0.3$, and $R = 10$.

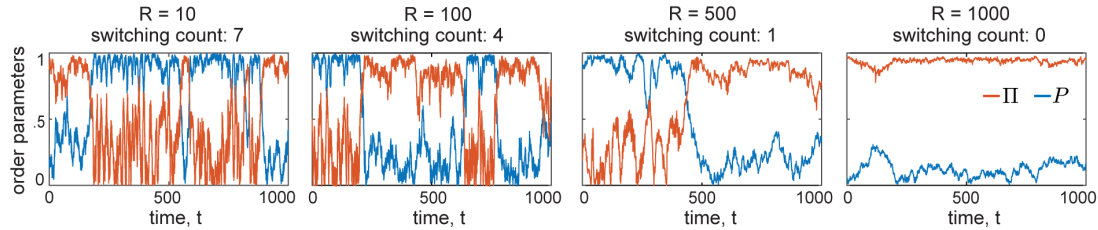


Figure S.9: Domain size affects intermittent bistability: Switching rate between milling and schooling decreases with increasing radius of the circular domain. That is, switching rate is larger in stronger confinement (smaller radius). This evident from the time evolution of the order parameters, here depicted for a time period of 1000 units selected after steady-state was reached. Parameter values are set to $N = 10$, $I_a = 1$, $I_n = 0.3$ and total integration time $T = 4000$.

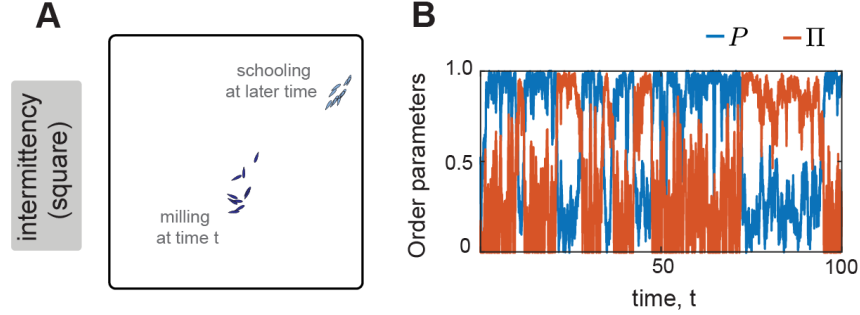


Figure S.10: **Robustness of intermittent behavior to domain geometry:** **A** Simulation of a group of $N = 10$ fish in a square geometry of side length $R = 10$ shows intermittent behavior between schooling and milling for a square domain of size comparable to the circular domain. **B** This is evident from the time evolution of the order parameters, here depicted for a time period of 100 units selected after steady-state was reached. Parameter values are set to $I_a = 1$ and $I_n = 0.3$.

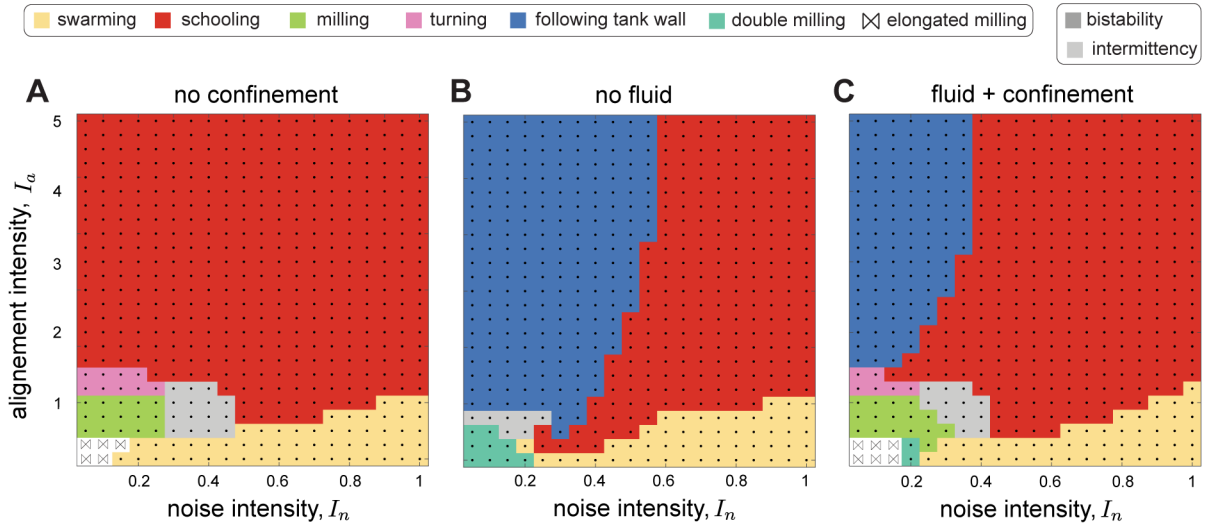


Figure S.11: **Phase diagrams** over the phase space (I_n, I_a) for $N = 10$: **A** Full model accounting for vision-based and hydrodynamic interactions in an unbounded domain. **B** Vision-based model without hydrodynamic interactions in geometric confinement. **C** Full model accounting for vision-based and hydrodynamic interactions under circular confinement. Distinct collective patterns are represented by their unique color code based on the value of the order parameters P and Π . Classification at each point (I_n, I_a) of the phase space is based on Monte-Carlo simulations consisting of 100 trials. In all simulations, $I_f = 10^{-2}$, $I_w = 0.94$, $R = 10$.

Algorithm S.1 Classification algorithm: multi-step classification algorithm for distinguishing collective behaviors based on the p.d.f.'s of P , Π , P_{wall} and Π_{mill} .

Algorithm: Classification of collective behavior

Input:

$$P = \frac{1}{N} \left\| \sum_{j=1}^N \mathbf{p}_j \right\|$$

$$P_{\text{wall}} = \frac{1}{N} \left| \sum_{j=1}^N \mathbf{p}_j \cdot \mathbf{t}_j \right|$$

$$\Pi = \frac{1}{N} \left\| \sum_{j=1}^N \frac{(\mathbf{r}_j - \mathbf{r}_c) \times \dot{\mathbf{r}}_j}{\|\mathbf{r}_j - \mathbf{r}_c\| \|\dot{\mathbf{r}}_j\|} \right\|$$

$$\Pi_{\text{mill}} = \frac{1}{N} \sum_{j=1}^N \frac{\|(\mathbf{r}_j - \mathbf{r}_c) \times \dot{\mathbf{r}}_j\|}{\|\mathbf{r}_j - \mathbf{r}_c\| \|\dot{\mathbf{r}}_j\|}$$

Bimodal distributions of P and Π :
calculate potential function

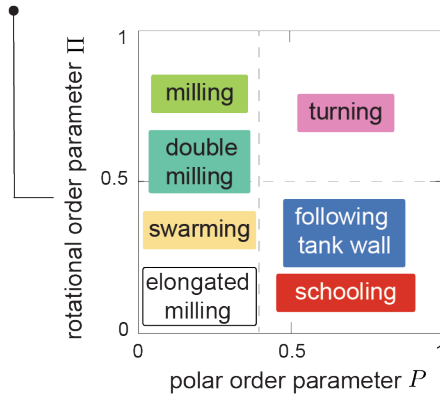
infinite
potential barrier

bistability

finite
potential barrier

intermittency

Unimodal distribution of P :
calculate time average values of P and Π



To distinguish swarming, milling and double milling:

Multimodal Π :

and Π_{mill} -average value > 0.5: double milling

Unimodal Π :

Π -average value > 0.5: milling

Π -average value < 0.5: swarming

elongated milling

To distinguish schooling and following tank wall:

calculate variance of P_{wall}

P_{wall} -variance > threshold: schooling

P_{wall} -variance < threshold: following tank wall

557 S.1 Dipolar Far-field Flow of Confined Swimmers

558 **Fish model.** Fish are modeled as planar self-propelled particles moving at constant
 559 speed U relative to the flow and exhibiting no inertia [21] (Fig. S.1A). We let (x, y)
 560 denote the Cartesian coordinates in the plane of motion. We introduce the inertial
 561 frame $(\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3)$, such that $(\mathbf{e}_1, \mathbf{e}_2)$ are unit vectors that span the x - and y -axes and
 562 $\mathbf{e}_3$ is a unit vector perpendicular to the (x, y) plane of motion. Considering N identical
 563 fish, the state of fish n ($n = 1, \dots, N$) is described by its position vector $\mathbf{r}_n \equiv (x_n, y_n)$
 564 expressed in the inertial frame $(\mathbf{e}_1, \mathbf{e}_2)$ and by its heading direction expressed as a unit
 565 vector $\mathbf{p}_n \equiv (\cos \theta_n, \sin \theta_n)$, where θ_n is the angle of the fish heading relative to the
 566 x -axis.

567 Let $\mathbf{u}$ be the fluid velocity field generated by the swimming fish, and let u_x and u_y
 568 be its components in the inertial frame, such that $\mathbf{u} \equiv (u_x, u_y)$. We seek an analytical
 569 expression of $\mathbf{u}$ based on the far-field flow generated by swimming fish.

570 **Dipolar far-field flow of a swimming fish.** A swimming fish generates a dipolar
 571 far-field flow [21, 26, 27] of strength $\kappa_f = UA$ that depends on the fish swimming
 572 speed U and cross-sectional surface area $A = \pi\ell^2/4$, where ℓ is the fish length. For
 573 convenience, we introduce the complex variable $z = x + iy$, where $i^2 = -1$, and we
 574 write the fluid velocity field in complex notation. For N fish located at $z_n = x_n + iy_n$ and
 575 headed at orientation angles θ_n , where $n = 1, \dots, N$, the complex potential function
 576 $F(z) = \phi + i\psi$ and the conjugate $w = u_x - iu_y = \frac{dF}{dz}$ of the fluid velocity field are given
 577 by superposition of N dipoles,

$$F(z) = -\sum_{n=1}^N \frac{\kappa_f e^{i\theta_n}}{\pi(z - z_n)}, \quad w(z) = \sum_{n=1}^N \frac{\kappa_f e^{i\theta_n}}{\pi(z - z_n)^2}. \quad (5)$$

Dipolar flow under geometric confinement. Under circular confinement, the dipolar flow field generated by a swimming fish should satisfy the zero penetration condition at the circular boundary (Fig. S.1B). The Milne-Thomson circle theorem states that for any irrotational flow with an associated complex potential function $F(z)$, analytic in the considered domain, a circular boundary of radius R can be constructed at the origin using the modified complex potential function $G(z) = F(z) + \overline{F(R^2/\bar{z})}$, where the overline denotes the complex conjugate (e.g., $\bar{z} = x - iy$). Accordingly, the velocity field generated by N fish confined in a circular domain is given by

$$w = u_x - iu_y = \frac{dG}{dz} = \sum_{n=1}^N \frac{\kappa_f e^{i\theta}}{\pi(z - z_n)^2} + \sum_{n=1}^N \frac{(\kappa_f R^2/z_n^2) e^{i(\pi-\theta_n)}}{\pi(z - R^2/\bar{z}_n)^2}. \quad (6)$$

The second term on the right-hand side of (6) corresponds to an image system, where for each dipole $(z_n, \theta_n; \kappa_f)$ inside the circle, an image dipole of intensity $\kappa_f R^2/z_n^2$ is located at a distance $R^2/\bar{z}_n$ and orientation $\pi - \theta_n$ (Fig. S.1B and [28]).

Regularization of the dipolar field. The point-dipole model suffers from singularities when fish get close to each other or to the confinement boundary. We regularize the point-dipole model by adding a regularization parameter δ . For N fish under circular confinement, the regularized counterpart of (6) is given by

$$w^\delta(z) = \sum_{n=1}^N \frac{\kappa_f e^{i\theta_n}}{\pi[(z - z_n)^2 + \delta^2]} + \sum_{n=1}^N \frac{\kappa_f R^2}{\pi \bar{z}_n^2} \frac{e^{i(\pi-\theta_n)}}{(z - R^2/\bar{z}_n)^2 + \delta^2 z^2/\bar{z}_n^2}. \quad (7)$$

In particular, the fluid velocity evaluated at the location of fish n is given by

$$w_n^\delta = \sum_{k=1, k \neq n}^N \frac{\kappa_f e^{i\theta_k}}{\pi[(z_n - z_k)^2 + \delta^2]} + \sum_{k=1}^N \frac{\kappa_f R^2}{\pi \bar{z}_k^2} \frac{e^{i(\pi-\theta_k)}}{(z_n - R^2/\bar{z}_k)^2 + \delta^2 z_n^2/\bar{z}_k^2}. \quad (8)$$

Equations of motion of hydrodynamically-coupled fish. Considering only hydrodynamic interactions among fish, a fish n moves under its own self-propelled velocity U in the θ_n direction, and it is also advected by the fluid velocity w_n^δ generated by other fish at the location z_n of fish n . By superposition, the rate of change of the fish position is given by

$$\dot{z}_n = Ue^{-i\theta_n} + w_n^\delta. \quad (9)$$

The fish also changes its heading in response to the fluid velocity gradient at z_n , which amounts to the difference in fluid velocity between the head and tail of fish [21, 27, 28]. Let Ω_n^h denote the rate of change of the heading of fish n due to hydrodynamic interactions, we have

$$\dot{\theta}_n = \Omega_n^h = \text{Re} \left[e^{i\theta_n} \frac{d\bar{w}_{\delta,n}}{dz} i e^{i\theta_n} \right]. \quad (10)$$

Here, $\text{Re}[\cdot]$ denotes the real part of the quantity if the square bracket. Substituting (8) into (9) and (10), we get

$$\dot{z}_n = Ue^{-i\theta_n} + \sum_{k=1, k \neq n}^N \frac{\kappa_f e^{i\theta_k}}{\pi [(z_n - z_k)^2 + \delta^2]} + \sum_{k=1}^N \frac{\kappa_f R^2}{\pi \bar{z}_k^2} \frac{e^{i(\pi - \theta_k)}}{(z_n - R^2/\bar{z}_k)^2 + \delta^2 z_n^2/\bar{z}_k^2}, \quad (11)$$

and

$$\begin{aligned} \Omega_n^h = \text{Re} \left[\sum_{k=1, k \neq n}^N -\frac{2\kappa_f i (z_n - z_k) e^{2i\theta_n} e^{i\theta_k}}{\pi [(z_n - z_k)^2 + \delta^2]^2} \right] + \\ \text{Re} \left[\sum_{k=1}^N \frac{2\kappa_f i R^2 e^{i2\theta_n} e^{i\theta_k}}{\pi \bar{z}_k^2} \frac{e^{i(\pi - \theta_k)} [(z_n - R^2/\bar{z}_k) + z_n \delta^2/\bar{z}_k]}{[(z_n - R^2/\bar{z}_k)^2 + \delta^2 z_n^2/\bar{z}_k^2]^2} \right]. \end{aligned} \quad (12)$$

S.2 Vision-based Response of Confined Swimmers

Following the behavioral rules in [18, 20, 21], a fish adjusts its orientation as follows: it is attracted to its Voronoi neighbors with intensity κ_p (units $\text{m}^{-1} \cdot \text{s}^{-1}$), tends to align with the same neighbors with intensity κ_v (units m^{-1}), and is subject to a rotational noise with standard deviation σ (units $\text{rad} \cdot \text{s}^{-1/2}$). In addition, it avoids collision with the tank wall with intensity κ_w (units $\text{m} \cdot \text{s}^{-1}$).

Fish-fish interactions. Consider a focal fish n and its interaction with a fish k , where $k \in \mathcal{V}_n$ and $\mathcal{V}_n$ is the set of Voronoi neighbors of fish n . To write down the evolution equations of the fish according to the attraction and alignment behavioral rules, it is convenient to define the following notation (Fig. S.1C)

$$d_{nk} = \|\mathbf{r}_k - \mathbf{r}_n\|, \quad \theta_{nk} = \cos^{-1} \left(\mathbf{p}_n \cdot \frac{(\mathbf{r}_k - \mathbf{r}_n)}{\|\mathbf{r}_k - \mathbf{r}_n\|} \right), \quad \phi_{nk} = \cos^{-1} (\mathbf{p}_n \cdot \mathbf{p}_k). \quad (13)$$

The attraction interaction depends on the relative distance d_{nk} and angle θ_{nk} between two fish. Given that fish n is not attracted towards a neighbor located behind it and does not turn in response to a fish that is right ahead, the change of heading of fish n in response to its attraction to neighbor k is taken to be proportional to $d_{nk} \sin \theta_{nk}$. The alignment interaction is characterized by its dependence on the difference in heading ϕ_{nk} between the two fish. To align with its neighbor, fish n changes its heading at a rate proportional to its swimming speed U and degree of misalignment with its neighbor, that is, at a rate proportional to $U \sin \phi_{nk}$.

By linear superposition, the rate of change Ω_n^V of the heading of fish n induced by it

625 attraction and alignment with its Voronoi neighbors $\mathcal{V}_n$ is given by

$$\Omega_n^{\mathcal{V}} = \frac{1}{|\mathcal{V}_n|} \sum_{k \in \mathcal{V}_n} (\kappa_v U \sin \phi_{nk} + \kappa_p d_{nk} \sin \theta_{nk}), \quad (14)$$

626 where κ_v and κ_p are constant parameters that denote the strength of the fish alignment
 627 and attraction response. We average this rate of rotation over the Voronoi neighbors
 628 using the weight $(1 + \cos \theta_{nk})$, which models continuously a rear blind angle of the fish
 629 visual field,

$$\Omega_n^{\mathcal{V}} = \left\langle \kappa_v U \sin \phi_{nk} + \kappa_p d_{nk} \sin \theta_{nk} \right\rangle, \quad (15)$$

630 where the bracket denotes the following nonlinear operation

$$\left\langle \circ \right\rangle = \frac{\sum_{k \in \mathcal{V}_n} \circ (1 + \cos \theta_{nk})}{\sum_{k \in \mathcal{V}_n} (1 + \cos \theta_{nk})}. \quad (16)$$

631 **Fish-wall interactions.** According to the behavioral rules inferred from shallow-water
 632 tracking experiments [18], to avoid collision with the tank wall, fish turn at the rate

$$\Omega_n^w = \kappa_w \frac{\text{sgn}(\phi_{nw})}{d_{nw}}. \quad (17)$$

633 Here, d_{nw} is the solution to the algebraic equation $\|\mathbf{r}_n + d_{nw} \mathbf{w} \mathbf{p}_n\| = R$ (this quadratic
 634 equation has two solutions: d_{nw} corresponds to the root that is in the field of view of the
 635 fish as dictated by the fish heading direction), and ϕ_{nw} is the angle given by

$$\phi_{nw} = \cos^{-1} \left(\mathbf{p}_n \cdot \frac{(\mathbf{r}_n + d_{nw} \mathbf{p}_n)}{\|\mathbf{r}_n + d_{nw} \mathbf{p}_n\|} \right). \quad (18)$$

636 **Fish free-will.** We model the fish free will, that is, its deviation from these strict behav-
 637 ioral rules, as a Wiener process. A Wiener process is a real-valued continuous-time

638 Markovian stochastic process: on a time interval $[0, T]$, it is a random variable $W(t)$
639 whose increment dW depends continuously on the time increment dt . Here, $\mathcal{N}(0, 1)$ is
640 a Gaussian (normal) distribution of zero mean and standard deviation equal to 1. In
641 our model, the Wiener process acts as noise in the rate of change of the fish heading
642 direction with a standard deviation σ ; we thus write it as σdW .

643 **Equations of motion of fish under vision-based rules.** Assuming no hydrody-
644 namic interactions, the motion of fish n is governed by the stochastic differential equa-
645 tions (SDE)

$$d\bar{z}_n = U e^{-i\theta_n} dt, \quad d\theta_n = (\Omega_n^{\mathcal{V}} + \Omega_n^{\mathcal{W}}) dt + \sigma dW. \quad (19)$$

646 Here, $\Omega_n^{\mathcal{V}}$ and $\Omega_n^{\mathcal{W}}$ are given by (15) and (17), respectively, and $dW = \sqrt{dt} \mathcal{N}(0, 1)$.

647 S.3 Equations of Motion

648 **Fish motion are governed by stochastic differential equations.** Considering the
649 combined effects of hydrodynamic interactions (11,12) and vision-based behavioral

	Complex notation	Vector notation
Translational velocity	$U e^{-i\theta_n}$ w_n^{δ}	$U \mathbf{p}_n$ $\mathcal{U} _{\mathbf{r}_n}$
Orientalional velocity	$\Omega_n^{\mathcal{V}} + \Omega_n^{\mathcal{W}}$ $\Omega_n^{\mathcal{H}}$	Ω_n $(\mathbf{p}_n \cdot \nabla) \mathcal{U} _{\mathbf{r}_n} \cdot \mathbf{p}_n^{\perp}$

Table S.2: **Notation:** translational and rotational velocity of swimmer n in complex and vector forms.

rules (19), the motion of fish n is governed by the SDEs

$$d\bar{z}_n = (Ue^{-i\theta_n} + w_n^\delta) dt, \quad d\theta_n = (\Omega_n^h + \Omega_n^\nu + \Omega_n^w)dt + \sigma dW. \quad (20)$$

Here, w_n^δ , Ω_n^h , Ω_n^ν , and Ω_n^w are given by (8), (12), (15), (17), and $dW = \sqrt{dt}\mathcal{N}(0, 1)$.

Dimensionless parameters. We introduce the time scale $\sqrt{1/(U\kappa_p)}$ and length scale $\sqrt{U/\kappa_p}$. The dimensionless counterparts of the fish speed U and attraction intensity κ_p are equal to 1, whereas the dimensionless counterparts of the dipolar field strength κ_f , alignment strength κ_v , wall-avoidance intensity κ_w , noise intensity σ are given by I_f , I_a , I_w , and I_n , respectively,

$$I_f = \kappa_f \kappa_p / U^2, \quad I_a = \kappa_v \sqrt{U/\kappa_p}, \quad I_w = \kappa_w / U, \quad I_n = \sigma (U\kappa_p)^{-1/4}. \quad (21)$$

We rewrite the stochastic differential equations (20) governing the fish motion in dimensionless form

$$\begin{aligned} d\bar{z}_n &= \left(e^{-i\theta_n} + \sum_{k=1, k \neq n}^N \frac{I_f e^{i\theta_k}}{\pi [(z_n - z_k)^2 + \delta^2]} + \sum_{k=1}^N \frac{I_f R^2}{\pi \bar{z}_k^2} \frac{e^{i(\pi - \theta_k)}}{(z_n - R^2/\bar{z}_k)^2 + \delta^2 z_n^2 / \bar{z}_k^2} \right) dt, \\ d\theta_n &= \left(\left\langle I_a \sin \phi_{nk} + d_{nk} \sin \theta_{nk} \right\rangle + I_w \frac{\text{sgn}(\phi_{nw})}{d_{nw}} + \right. \\ &\quad \left. \text{Re} \left[\sum_{k=1, k \neq n}^N -\frac{2I_f i(z_n - z_k) e^{2i\theta_n} e^{i\theta_k}}{\pi [(z_n - z_k)^2 + \delta^2]^2} \right] + \right. \\ &\quad \left. \text{Re} \left[\sum_{k=1}^N \frac{2I_f i R^2 e^{i2\theta_n} e^{i\theta_k}}{\pi \bar{z}_k^2} \frac{e^{i(\pi - \theta_k)} [(z_n - R^2/\bar{z}_k) + z_n \delta^2 / \bar{z}_k]}{[(z_n - R^2/\bar{z}_k)^2 + \delta^2 z_n^2 / \bar{z}_k^2]^2} \right] \right) dt + I_n dW. \end{aligned} \quad (22)$$

Rewriting equations of motion in vector form. We rewrite (20) in vector notation in terms of the position vector $\mathbf{r}_n$ and heading direction $\mathbf{p}_n$ of swimmer n (notation

661 summarized in Table S.2)

$$\begin{aligned}\dot{\mathbf{r}}_n &= \mathbf{p}_n + \mathcal{U}|_{\mathbf{r}_n}, \\ d\theta_n &= [\Omega_n + (\mathbf{p}_n \cdot \nabla)\mathcal{U}|_{\mathbf{r}_n} \cdot \mathbf{p}_n^\perp] dt + I_n dW.\end{aligned}\tag{23}$$

662 Here, the heading direction is written in terms of the unit vector $\mathbf{p}_n = [\cos \theta_n, \sin \theta_n]$,
 663 $\mathcal{U}|_{\mathbf{r}_n}$ is a vector representing the fluid velocity at the location of swimmer n , namely,
 664 $\mathcal{U}|_{\mathbf{r}_n} \equiv (\text{Re}[w_n^\delta], -\text{Im}[w_n^\delta])$, $\Omega_n = \Omega_n^v + \Omega_n^w$ is a scalar representing the vision-based
 665 rules including fish-fish interactions and fish-wall interactions. The hydrodynamic term
 666 Ω_n^h is written in vector form as $(\mathbf{p}_n \cdot \nabla)\mathcal{U}|_{\mathbf{r}_n} \cdot \mathbf{p}_n^\perp$.

667 **Numerical integration.** We integrate (22), or equivalently, (23), numerically using the
 668 Euler-Maruyama integration algorithm for stochastic differential equations [39]. For a
 669 uniform discretization of the time interval $[0, T]$ using a fixed time step Δt , the stochastic
 670 equations are discretized using a straightforward Euler approximation, where the noise
 671 term dW is approximated using $\Delta W_t = W_{t+\Delta} - W_t$. The increments ΔW_t are sampled
 672 from an independent Gaussian random variable with zero mean and variance of Δt .

673 S.4 Group Dynamics and the Fokker-Planck Equation

674 **Fokker-Planck formulation.** We consider that the system dynamics at the macro-
 675 scopic group level can be characterized by statistical order parameters. We con-
 676 sider describing the evolution of the statistical order parameters as a continuous m -
 677 dimensional random variable $\mathbf{X}(t) \equiv (P, \Pi, \dots)$ governed by the multivariate Langevin

equation [24, 31],

$$dX_i(t) = A_i(\mathbf{X}(t), t)dt + \sum_{j=1}^M b_{ij}(\mathbf{X}(t), t)\Gamma_j(t)dt, \quad (24)$$

where the functions A_i and b_{ij} are all smooth, and $\Gamma_j(t)$ represents the fluctuating Langevin forces associated with a Gaussian white noise,

$$\Gamma_j(t) \equiv \lim_{dt \rightarrow 0} \mathcal{N}(0, 1/dt). \quad (25)$$

The statistically independent, temporal uncorrelated nature of $\Gamma_j(t)$ implies that the ensemble average $\langle \Gamma_j(t)\Gamma_k(t') \rangle$ vanishes if either $j \neq k$ or $t \neq t'$.

We introduce the probability density function $p(\mathbf{x}, t)$ such that $p(\mathbf{x}, t|\mathbf{x}_0, t_0)d\mathbf{x}$ denotes the probability of finding $\mathbf{X}(t)$ between $\mathbf{x}$ and $\mathbf{x} + d\mathbf{x}$, given that $\mathbf{X}(t_0) = \mathbf{x}_0$ for $t \geq t_0$. Then, the probability density function p satisfies the Fokker-Planck partial differential equation

$$\begin{aligned} \frac{\partial p(\mathbf{x}, t|\mathbf{x}_0, t_0)}{\partial t} = & - \sum_{i=1}^M \frac{\partial}{\partial x_i} [V_i(\mathbf{x}, t)p(\mathbf{x}, t|\mathbf{x}_0, t_0)] + \\ & \frac{1}{2} \sum_{i=1, j=1}^M \frac{\partial^2}{\partial x_i \partial x_j} [D_{ij}(\mathbf{x}, t)p(\mathbf{x}, t|\mathbf{x}_0, t_0)] \end{aligned} \quad (26)$$

where V_i and D_{ij} are drift and diffusion coefficients, respectively, that can be related to the random terms in (24),

$$V_i(\mathbf{x}, t) = A_i(\mathbf{x}, t), \quad D_{ij}(\mathbf{x}, t) = \sum_{k=1}^M b_{ik}(\mathbf{x}, t)b_{jk}(\mathbf{x}, t). \quad (27)$$

For the class of stationary continuous Markovian processes with uncorrelated noise,

690 drift and diffusion coefficients have statistical definitions from time series data [24]:

$$V_i(\mathbf{x}, t) = \lim_{\tau \rightarrow 0} \frac{1}{\tau} \langle X_i|_{t+\tau} - x_i \rangle \Big|_{\mathbf{X}(t)=\mathbf{x}} \quad (28)$$

691

$$D_{ij}(\mathbf{x}, t) = \lim_{\tau \rightarrow 0} \frac{1}{\tau} \langle (X_i|_{t+\tau} - x_i)(X_j|_{t+\tau} - x_j) \rangle \Big|_{\mathbf{X}(t)=\mathbf{x}} \quad (29)$$

692 **One-dimensional Fokker-Planck equation and the effective potential formulation.**

693 For a one-dimensional variable, the univariate Fokker-Planck equation can be written

694 as

$$\frac{\partial p(x, t)}{\partial t} = -\frac{\partial}{\partial x} [V(x)p(x, t)] + \frac{\partial^2}{\partial x^2} [D(x)p(x, t)]. \quad (30)$$

695 The stationary distribution ($\partial p(x, t)/\partial t = 0$) of this equation has an analytical solution

696 in terms of an effective potential $\Phi(x)$. Stationary solutions of (30) satisfy

$$\frac{\partial}{\partial x} \left[V(x)p(x, t) - \frac{\partial D(x)p(x, t)}{\partial x} \right] = 0 \quad (31)$$

697 Hence,

$$p(x, t) = \frac{C}{D(x)} \exp \left(\int_0^x \frac{V(x')}{D(x')} dx' \right). \quad (32)$$

698 where C is a normalization constant. Defining

$$p(x, t) \sim \exp(-\Phi(x)), \quad (33)$$

699 it follows that $\Phi(x)$ satisfies

$$\Phi(x) = \ln(D(x)) - \int_0^x \frac{V(x')}{D(x')} dx' + \text{constant}. \quad (34)$$

Dimensionality reduction. To study the transitional dynamic of the fish group, we consider $\mathbf{X}(t) \equiv (P(t), \Pi(t))$. Observations of the bivariate distribution of $(P(t), \Pi(t))$ based on numerical simulations in the bistable and intermittent regimes indicate that the data lies predominantly on a lower one-dimensional space in the $(P(t), \Pi(t))$ plane. We thus seek a dimensionality reduction from $(P(t), \Pi(t))$ to a one-dimensional space.

We first construct the data matrix $\mathcal{M}$, where each row represents time step and each column represents a feature, and subtract the mean value of each column.

$$\mathcal{M} = \begin{bmatrix} P_{t_0=0} - \langle P \rangle & \Pi_{t_0=0} - \langle \Pi \rangle \\ \vdots & \vdots \\ P_{t_n=T} - \langle P \rangle & \Pi_{t_n=T} - \langle \Pi \rangle \end{bmatrix}. \quad (35)$$

Our two-dimensional data is best reduced to a non-linear curve in the 2D space (P, Π) . Specifically, we choose a circular kernel and we seek a radius Θ of the reduced circular curve that minimizes the projection distance from the data points to that curve,

$$\underset{\Theta}{\operatorname{argmin}} \sum (\Theta^2 - \operatorname{tr}(\mathcal{M}\mathcal{M}^T)). \quad (36)$$

We solve (36) numerically to obtain the value of Θ . Once we have Θ , we project our data points onto the quarter-circle of radius Θ in the positive (P, Π) quadrant. After projection, the data on a quarter-circle of radius Θ can be described by a new coarse-grained variable defined as $K = (2/\pi) \tan^{-1}(P/\Pi)$, where $K = 0$ corresponds to milling ($P = 0$) and $K = 1$ corresponds to schooling ($\Pi = 0$).

Construction of drift and diffusion coefficients from numerical data. Following data reduction to the 1D circular curve, the reduced coarse-grained data is parame-

717 terized by the random variable $K(t)$. Considering that the data series $K(t)$ can be
 718 obtained from a 1D Langevin equation of the form (24), the associated probability den-
 719 sity function $p(x, t)$ can be expressed as a univariate Fokker-Planck equation (30),
 720 where the drift and diffusion coefficients V and D can be approximated numerically,
 721 following(28) and (29), from the time series of $K(t)$. Specifically, we first discretized
 722 $K(t)$ into K_t over a grid of values. Then, for every appearance of the grid value in the
 723 time series, with a certain error tolerance, we recorded the subsequent values over a
 724 short fixed time interval τ , and computed the ensemble mean $\langle K(t; x_n) \rangle$ and variance
 725 $\sigma^2(t; x_n)$ over these short trajectories. Finally, we estimated $V(x)$ and $D(x)$ using the
 726 slope of the linear regression of $\langle X(t; x_n) \rangle$ and $\sigma^2(t; x_n)$.

727 **S.5 The Fokker-Planck Equation and Kramer's Law**

728 **Derivation of the Fokker-Planck equation from the Langevin equation.** Our basic
 729 assumption is that the projected order parameter K follows the Langevin equation:

$$dK = V dt + \sigma dW_t, \quad (37)$$

730 where $V(K)$ is the drift and $\sigma(K)$ is the diffusion for the system, and both are functions
 731 of the random process K . Consider a continuous test function $f(K, t)$, and take the
 732 Taylor series expansion of f ,

$$df = \frac{\partial f}{\partial t} dt + \frac{1}{2} \frac{\partial^2 f}{\partial t^2} dt^2 + \cdots + \frac{\partial f}{\partial K} dK + \frac{1}{2} \frac{\partial^2 f}{\partial K^2} dK^2 + \cdots \quad (38)$$

733 Now substitute the Langevin equation into this expansion, ignoring higher order terms,

$$df = \left[\frac{\partial f}{\partial t} + V(K) \frac{\partial f}{\partial K} \right] dt + \sigma(K) \frac{\partial f}{\partial K} dW_t + \frac{1}{2} \sigma^2(K) \frac{\partial^2 f}{\partial K^2} dW_t^2. \quad (39)$$

734 Because $dW_t = \mathcal{N}(0, 1) \sqrt{dt}$ is independent of K , where $\mathcal{N}(0, 1)$ is a normal distri-
735 bution of zero mean and variance 1, we have by the rule of product of expectations,

$$736 \left\langle \sigma \frac{\partial f}{\partial K} dW_t \right\rangle = \left\langle \sigma \frac{\partial f}{\partial K} \right\rangle \langle dW_t \rangle = 0 \text{ and } \left\langle \frac{1}{2} \sigma^2 \frac{\partial^2 f}{\partial K^2} dW_t^2 \right\rangle = \left\langle \frac{1}{2} \sigma^2 \frac{\partial^2 f}{\partial K^2} \right\rangle \langle dW_t^2 \rangle = \left\langle \frac{1}{2} \sigma^2 \frac{\partial^2 f}{\partial K^2} \right\rangle dt.$$

737 Taking the expectation of the above equation and re-arranging terms, we get

$$\langle df \rangle = \left[\left\langle \frac{\partial f}{\partial t} \right\rangle + \left\langle V(K) \frac{\partial f}{\partial K} \right\rangle + \left\langle \frac{1}{2} \sigma^2(K) \frac{\partial^2 f}{\partial K^2} \right\rangle \right] dt. \quad (40)$$

738 At steady state, we have $\frac{df}{dt} = 0$,

$$0 = \left\langle \frac{\partial f}{\partial t} \right\rangle + \left\langle V(K) \frac{\partial f}{\partial K} \right\rangle + \left\langle \frac{\sigma^2(K)}{2} \frac{\partial^2 f}{\partial K^2} \right\rangle. \quad (41)$$

739 Writing the expectations explicitly in terms of the probability $p(K, t)$ of being in a state
740 K at time t ,

$$0 = \int_{\mathbb{R}} dt \left[\int_{\mathbb{R}} dK \frac{\partial f}{\partial t} p(K, t) + \int_{\mathbb{R}} dK V(K) \frac{\partial f}{\partial K} p(K, t) + \int_{\mathbb{R}} dK \frac{\sigma^2(K, t)}{2} \frac{\partial^2 f}{\partial K^2} p(K, t) \right] \quad (42)$$

741 Using integration by part, assuming proper decay of probabilities at infinity, we get

$$0 = - \int_{\mathbb{R}} dK f \frac{\partial p(K, t)}{\partial t} + \int_{\mathbb{R}} dK f \left[- \frac{\partial}{\partial K} \left(V(K) p(K, t) \right) + \frac{1}{2} \frac{\partial^2}{\partial K^2} \left(\sigma^2(K) p(K, t) \right) \right] \quad (43)$$

742 Since the function f is arbitrary, we have

$$0 = - \frac{\partial p(K, t)}{\partial t} - \frac{\partial}{\partial K} \left(V(K) p(K, t) \right) + \frac{1}{2} \frac{\partial^2}{\partial K^2} \left(\sigma^2(K) p(K, t) \right) \quad (44)$$

743 Introducing the notation $D(K) = \frac{1}{2}\sigma^2(K)$, we arrive at the Fokker-Planck equation,
 744 where both drift $V(K)$ and diffusion $D(K)$ depend on K ,

$$\frac{\partial p(K, t)}{\partial t} = -\frac{\partial}{\partial K} [V(K)p(K, t)] + \frac{\partial^2}{\partial K^2} [D(K)p(K, t)] \quad (45)$$

745 We write the Fokker-Planck equation in terms of the flux $J(K, t)$ of probability

$$\frac{\partial p}{\partial t} + \frac{\partial J}{\partial K} = 0, \quad \text{where} \quad J = V(K)p(K, t) - \frac{\partial D(K)p(K, t)}{\partial K}. \quad (46)$$

746 For stationary solutions with $\partial p/\partial t = 0$ and $J = 0$, we get, by rearranging terms and
 747 integrating with respect to K , the stationary probability density function $p_s(K)$,

$$0 = V(K)p_s(K) - \frac{\partial}{\partial K} [D(K)p_s(K)] \implies p_s(K) = \frac{C}{D(K)} \exp\left(\int_0^K \frac{V(K')}{D(K')} dK'\right). \quad (47)$$

748 Introducing the effective potential $\Phi(K)$, we write $p_s(K) \sim \exp(-\Phi(K))$.

749 **Classic Fokker Planck formulation and Kramer's Law.** Consider a system of over-
 750 damped particles subject to a potential force field $F(K) = -\partial\Psi/\partial K$, with associated
 751 potential function $\Psi(K)$, and undergoing Brownian motion at a given temperature T and
 752 thermal energy $k_B T$, where k_B is the Boltzman constant, with constant diffusivity coeffi-
 753 cient D . For overdamped systems, the particle velocity $V = -F/\gamma = -(1/\gamma)(\partial\Psi/\partial K)$,
 754 where γ is the friction coefficient, which is related to the diffusivity D by Einstein's
 755 relation $\gamma = k_B T/D$. The corresponding Fokker-Planck equation is given by

$$\frac{\partial p(K, t)}{\partial t} = \frac{\partial}{\partial K} \underbrace{\left(\frac{D}{k_B T} \frac{\partial \Psi}{\partial K} p(K, t) + D \frac{\partial p(K, t)}{\partial K} \right)}_{-J(K)} \quad (48)$$

756 Noting that $\frac{\partial}{\partial K} \left(\exp\left(\frac{\Psi}{k_B T}\right) \right) = \frac{1}{k_B T} \frac{\partial \Psi}{\partial K} \exp\left(\frac{\Psi}{k_B T}\right)$, rearranging and substituting into the
 757 expression of the probability flux to get

$$J = -D \exp\left(-\frac{\Psi}{k_B T}\right) \frac{\partial}{\partial K} \left[\exp\left(\frac{\Psi}{k_B T}\right) p \right]. \quad (49)$$

758 Using (49), we rewrite (48) in the form

$$\frac{\partial p(K, t)}{\partial t} = D \frac{\partial}{\partial K} \left\{ \exp\left(-\frac{\Psi}{k_B T}\right) \frac{\partial}{\partial K} \left[\exp\left(\frac{\Psi}{k_B T}\right) p \right] \right\}. \quad (50)$$

759 When the system is in equilibrium, the probability flux J in (49) is zero and we get
 760 $p = p_o \exp\left(-\frac{\Psi}{k_B T}\right)$ where p_o can be determined from normalization. However, such state
 761 of absolute equilibrium is not achievable in a Brownian system. Instead we assume that
 762 the system reaches a quasi-stationary state near the minimum of a potential well before
 763 the particle escapes.

764 Consider a double well potential with states $K_a = a$ and $K_c = c$ denoting the quasi-
 765 stationary states near the minimum of each well. Let $K_b = b$ denote the barrier top. The
 766 escape rate r is determined by the normalized flux of probability across the potential
 767 barrier

$$r = \frac{\text{Flux across the potential barrier}}{\text{Probability of being inside well}} = \frac{J}{P} \quad (51)$$

768 Considering that the initial state is near K_a inside well, we want to compute the rate r
 769 of escaping from K_a to K_c , which by definition is the normalized flux of probability from
 770 state K_a to state K_c .

771 To determine the probability flux from K_a to K_c , we start from (49) and rearrange

772 terms

$$\frac{\partial}{\partial K} \left(\exp \left(\frac{\Psi}{k_B T} \right) p \right) = -\frac{J}{D} \exp \left(-\frac{\Psi}{k_B T} \right). \quad (52)$$

773 Integrating between points $K_a = a$ and $K_c = c$, recalling that initially, the probability of
774 being at K_c is nearly zero, we get

$$\exp \frac{\Psi(K)}{k_B T} p \Big|_a^c = -\exp \left(\frac{\Psi(a)}{k_B T} \right) p(a) = -\frac{J}{D} \int_a^c \exp -\frac{\Psi(K')}{k_B T} dK', \quad (53)$$

775 Therefore the flux across the barrier from K_a to K_c is given by

$$J = \frac{D \exp \frac{\Psi(a)}{k_B T} p(a)}{\int_a^c \exp -\frac{\Psi(K')}{k_B T} dK'}. \quad (54)$$

776 The denominator is maximum about the top of the potential barrier $K_b = b$. Expanding
777 about this point, we get

$$\begin{aligned} \int_a^c \exp -\frac{\Psi(K')}{k_B T} dK' &= \int_a^c \exp \left(-\frac{\Psi(b) + \Psi''(b)(K' - b)^2/2 + \dots}{k_B T} \right) dK' \\ &\approx \exp \left(\frac{\Psi(b)}{k_B T} \right) \int_{-\infty}^{\infty} \exp \left(-\frac{|\Psi''(b)|(K' - b)^2}{2k_B T} \right) dK' \\ &= \left(\frac{2\pi k_B T}{|\Psi''(b)|} \right)^{1/2} \exp \left(\frac{\Psi(b)}{k_B T} \right) \end{aligned} \quad (55)$$

778 To evaluate the probability P of being near K_a inside the well, we assume the barrier
779 is high, so that we can employ the equilibrium relation, $p(K) = p(a) \exp \left(-\frac{\Psi(K) - \Psi(a)}{k_B T} \right)$.

780 We get

$$P = \int_{a-\Delta}^{a+\Delta} p(K) dK = p(a) \exp \frac{\Psi(a)}{k_B T} \int_{a-\Delta}^{a+\Delta} \exp -\frac{\Psi(K)}{k_B T} dK \quad (56)$$

where Δ is of the order of the size of the well. The integrand is maximum around the state $K_a = a$; expanding about this state gives

$$P = p(a) \left(\frac{2\pi k_B T}{|\Psi''(a)|} \right)^{1/2} \quad (57)$$

Substituting both results for J and P into (51), we get Kramer's law for the escape rate

$$r = \frac{D}{2\pi k_B T} \left(|\Psi''(a)| |\Psi''(b)| \right)^{1/2} \exp \left(-\frac{\Psi(b) - \Psi(a)}{k_B T} \right) \quad (58)$$

In this derivation, the diffusion coefficient D is constant; it is not a function of state K .

Distinction between our system and the classic system of overdamped particles

subject to Brownian noise. There are multiple differences between our system and the system of overdamped particles subject to Brownian noise. A major difference is that the latter admits a well-defined temperature T and thermal energy $k_B T$, while the "temperature" in our system is hidden and arises from disparate sources of randomness due continuous rotational noise at the individual level and due to initial conditions and Monte Carlo simulations. The "temperature" or "thermal energy" associated with these sources of randomness is not explicit. In our system, both the effective drift and diffusion coefficients are function of the random process K , whereas in the classic set-up for which Kramer's law was derived, diffusion D is constant. Also, in our formulation, the effective potential Φ includes all sources of randomness, whereas the potential function Ψ in the classic system accounts only for deterministic drift.

In an effort to write our system in terms of a potential function Ψ that, unlike the effective potential Φ , does not account for "thermal energy," we start from the probability

flux J at steady state and introducing Ψ such that $d\Psi/dK = -(V - \partial D/\partial K)$, we have

$$J = Vp_s - p_s \frac{\partial D}{\partial K} - D \frac{\partial p_s}{\partial K} = -\frac{\partial \Psi}{\partial K} p_s - D \frac{\partial p_s}{\partial K} = -\frac{\partial \Psi}{\partial K} (p_s + D \frac{\partial p_s}{\partial \Psi}) = 0$$

Rearranging terms and integrating both sides, we get

$$p_s + D \frac{\partial p_s}{\partial \Psi} = 0 \quad \Rightarrow \quad \frac{dp_s}{p_s} = -\frac{d\Psi}{D} \quad \Rightarrow \quad p_s = p_0 \exp\left(-\frac{\Psi}{D}\right)$$

797 This is analogous to the classic system where $p_s \sim \exp(-\frac{\Psi}{k_B T})$, but in our system

798 $D(K) = k_B T$ implies that the temperature T is not constant.